Out-of-Sample Cross-Validation of the RCV Support Models

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```r
library(dplyr)
library(knitr)
library(kableExtra)
library(ggplot2)
library(tidyr)
library(RColorBrewer)

load("../data/cv_results.RData")   # cv_results (from R/05_cv.R)

metrics      <- cv_results$metrics
fold_metrics <- cv_results$fold_metrics
fold_mean    <- cv_results$fold_mean
imp_spread   <- cv_results$imp_spread
prov         <- cv_results$provenance

# convenience: a named getter for headline numbers used in prose
m1 <- metrics[1, ]; m2 <- metrics[2, ]; m3 <- metrics[3, ]
pp <- function(x) sprintf("%.1f", 100 * x)   # proportion -> percentage-point string
```

Why this report exists

The models in `04_models.R` report an in-sample adjusted pseudo-R^2 of **0.86** (baseline) and **0.83** (+ demographics). Those numbers measure fit to the *same* precincts the models were trained on. They cannot tell a reader how well the model would predict RCV support in a **jurisdiction it has never seen** — which is the only claim worth putting on a résumé. This report replaces the in-sample number with a genuinely out-of-sample one.

> **Headline:** under leave-one-locale-out cross-validation, the baseline model `dem_share * recent_lpw` predicts a held-out jurisdiction's precinct-level RCV

support with a mean absolute error of **7.9 percentage points** (out-of-sample deviance-R^2 = **+0.57**), pooling across 5 imputations. Every number below is out-of-sample.

Design decisions

Grouping: leave-one-locale-out (LOLO), 8 folds. Precincts are nested within 8 locales, and locale drives almost everything — the outcome, the key regressor `recent_lpw` (a locale-level constant), and M3's random intercept. A naive random precinct split would let the model learn each locale's mean from its own training precincts and then "predict" its held-out precincts, leaking exactly the cluster structure we want to test across. Each fold therefore holds out one **entire** locale and predicts it from the other seven. This is the correct — and deliberately harsh — test of generalisation to a new jurisdiction. It can honestly produce a **negative** deviance-R^2 when a model does worse than a training-mean intercept.

Only Alaska and Maine have `recent_lpw = TRUE`, so LOLO keeps `recent_lpw` identified in every fold (holding out one leaves the other). Holding out Alaska or Maine is thus a direct out-of-sample test of the paper's central LPW hypothesis: *does the LPW shift estimated from Maine alone actually predict Alaska, and vice versa?*

Imputation under CV (leakage-aware).

- **M1** uses only `dem_share * recent_lpw` — no demographics, so **no imputation**. Its CV is fully leakage-free. This is the cleanest headline number.
- **M2 / M3** use six Census demographics with missing values. The `mice` imputation model is **refit inside each training fold** via mice's `ignore=` argument: the held-out locale's rows are imputed but never contribute to estimating the imputation model, so no held-out information leaks into the imputed predictors. *Documented compromise:* under LOLO the held-out locale is an unseen factor level, so we drop `locale` from the imputation predictors (03_imputation.R used it) and impute demographics from `dem_share`, `recent_lpw`, and the other demographics. Held-out imputation therefore regresses toward training patterns — the honest reality of predicting a new jurisdiction's missing demographics.

Metrics (appropriate to a quasibinomial / proportion outcome):

- **OOS deviance-explained** — `1 - Σ deviance(model) / Σ deviance(null)`, null = each fold's training-set overall support rate, pooled over all held-out precincts. The out-of-sample analog of the in-sample pseudo-R^2.
- **MAE / RMSE** of the predicted support share (percentage points), precinct-level and vote-weighted.

All metrics pool across the 5 imputations by averaging the predicted probabilities per precinct, then scoring once.

Pooled out-of-sample results

```
metrics %>%
  transmute(
    Model = model,
    `OOS dev-R²` = round(oos_dev_r2, 3),
    `MAE (pp)`   = round(100 * mae, 2),
    `RMSE (pp)`  = round(100 * rmse, 2),
    `Vote-wt MAE (pp)` = round(100 * mae_vote_wt, 2),
    `N held-out precincts` = n_precincts
  ) %>%
  kable(caption = paste0("Leave-one-locale-out metrics, pooled across ",
                         prov$m_imputations, " imputations. Every precinct scored ",
                         "only when its locale was held out.")) %>%
  kable_styling(full_width = FALSE)
```

Table 1: Leave-one-locale-out metrics, pooled across 5 imputations. Every precinct scored only when its locale was held out.

Model	OOS dev-R^2	MAE (pp)	RMSE (pp)	Vote-wt MAE (pp)	N he
M1 baseline (dem_share * recent_lpw)	0.569	7.85	10.05	7.37	
M2 + demographics	0.517	8.69	11.01	7.81	
M3 GLMM (+ demographics)	0.238	10.73	12.95	10.50	

Two ways of averaging tell slightly different stories: **precinct-pooled** (above, dominated by the large Massachusetts and Alaska folds) versus **fold-mean** (each of the 8 locales weighted equally, so the tiny cities count as much as Massachusetts).

```
metrics %>%
  select(model, oos_dev_r2, mae) %>%
  left_join(
    fold_mean %>%
      mutate(model = metrics$model[match(model, c("M1","M2","M3"))]),
    by = "model"
  ) %>%
  transmute(
    Model = model,
    `Dev-R² (precinct-pooled)` = round(oos_dev_r2, 3),
    `Dev-R² (fold-mean)`       = round(dev_r2_fold_mean, 3),
    `MAE pp (precinct-pooled)` = round(100 * mae, 2),
```

```
    `MAE pp (fold-mean)`        = round(100 * mae_fold_mean, 2)
  ) %>%
  kable(caption = "Precinct-pooled vs equal-weight-per-locale.") %>%
  kable_styling(full_width = FALSE)
```

Table 2: Precinct-pooled vs equal-weight-per-locale.

Model	Dev-R^2 (precinct-pooled)	Dev-R^2 (fold-mean)	MAE pp (precinct
M1 baseline (dem_share * recent_lpw)	0.569	0.425	
M2 + demographics	0.517	0.421	
M3 GLMM (+ demographics)	0.238	0.774	

Per-held-out-locale detail

Which jurisdictions are hard to predict from the others? This is the table that makes the LOLO test concrete.

```
fold_metrics %>%
  filter(model == "M1") %>%
  transmute(
    `Held-out locale` = locale,
    `N precincts` = n_precincts,
    `OOS dev-R²` = round(oos_dev_r2, 3),
    `MAE (pp)`   = round(100 * mae, 2),
    `Mean observed support` = round(mean_obs, 3),
    `Mean predicted`        = round(mean_pred, 3)
  ) %>%
  kable(caption = "M1 baseline, one row per held-out locale.") %>%
  kable_styling(full_width = FALSE)
```

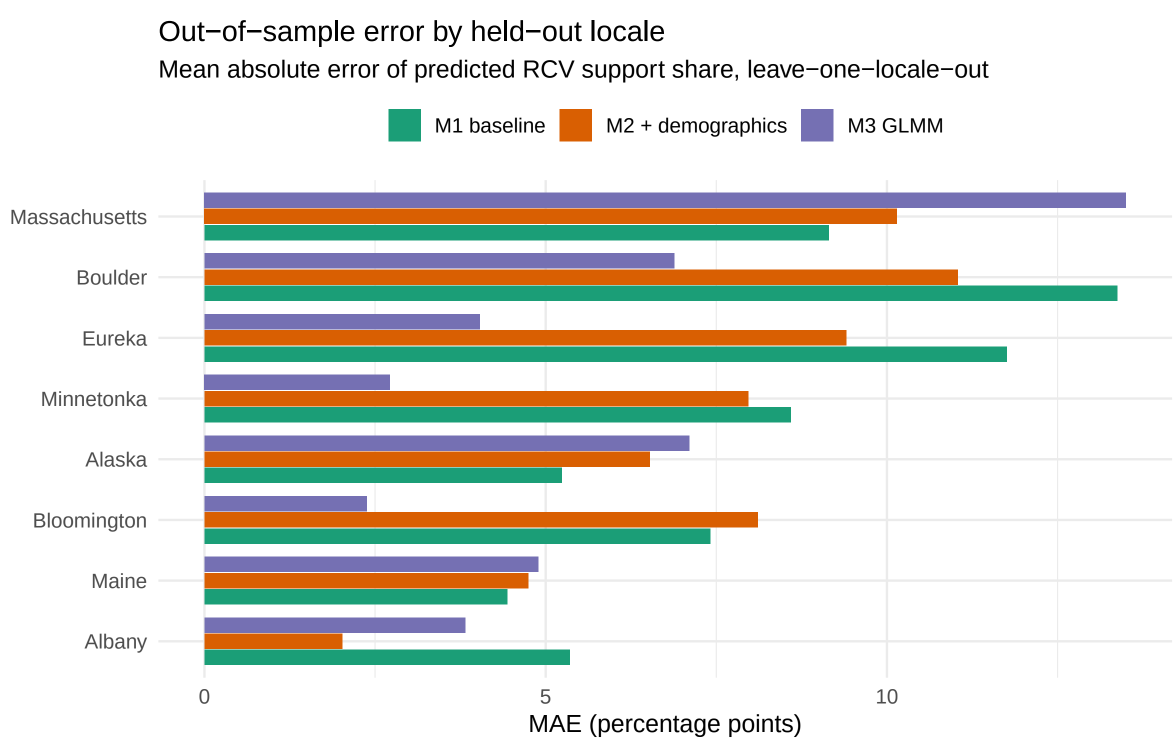
Table 3: M1 baseline, one row per held-out locale.

Held-out locale	N precincts	OOS dev-R^2	MAE (pp)	Mean observed support	Mean predicted
Alaska	578	0.908	5.23	0.486	0.483
Albany	3	0.943	5.35	0.733	0.680
Bloomington	32	-0.761	7.41	0.512	0.438
Boulder	88	0.789	13.38	0.782	0.648
Eureka	8	0.279	11.75	0.603	0.486

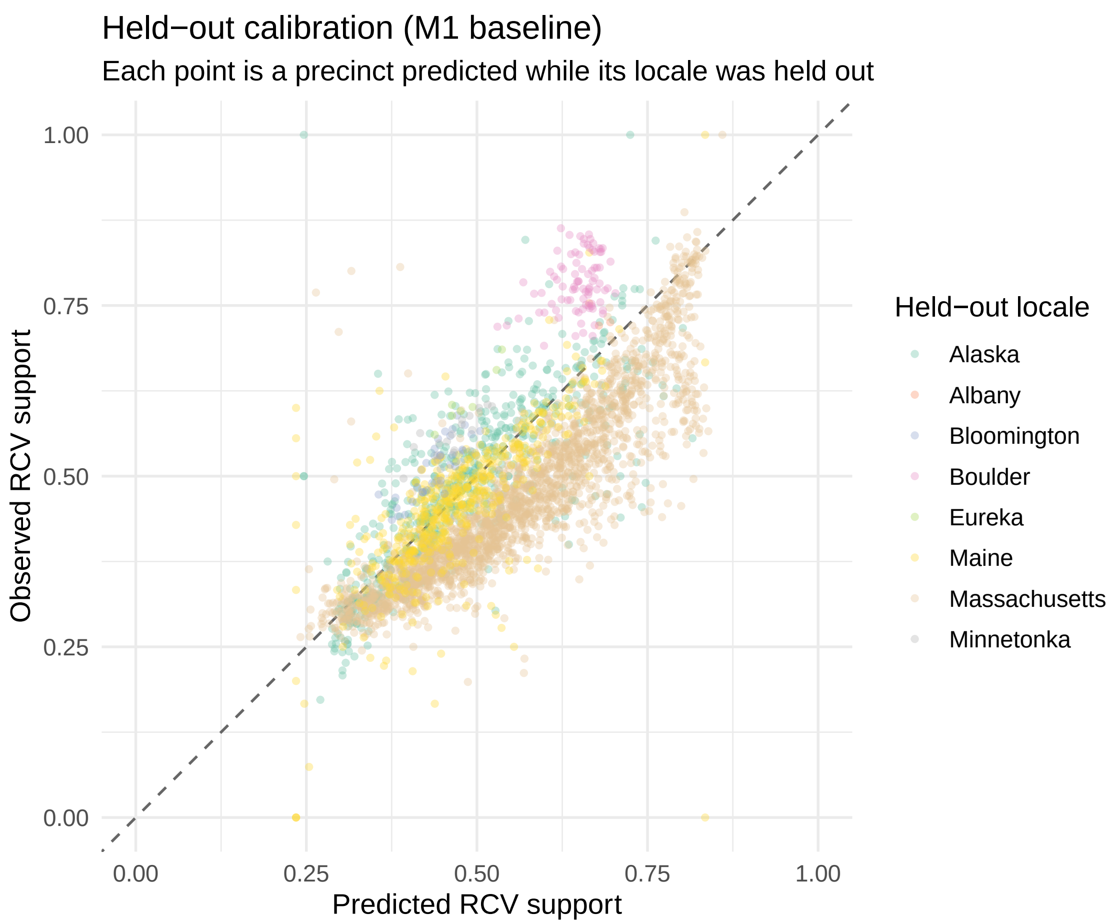
Maine	527	0.917	4.43	0.456	0.468
Massachusetts	2173	0.492	9.15	0.470	0.556
Minnetonka	23	-0.163	8.59	0.547	0.461

```r
plot_df <- fold_metrics %>%
  mutate(model = recode(model, M1 = "M1 baseline",
                        M2 = "M2 + demographics", M3 = "M3 GLMM"))

ggplot(plot_df, aes(x = reorder(locale, mae), y = 100 * mae, fill = model)) +
  geom_col(position = position_dodge(width = 0.8), width = 0.75) +
  scale_fill_brewer(palette = "Dark2", name = NULL) +
  coord_flip() +
  labs(
    title = "Out-of-sample error by held-out locale",
    subtitle = "Mean absolute error of predicted RCV support share, leave-one-locale-out",
    x = NULL, y = "MAE (percentage points)"
  ) +
  theme_minimal(base_size = 12) +
  theme(legend.position = "top")
```



```r
# Observed vs predicted support, held-out precincts (M1), coloured by locale.
cal <- cv_results$pooled_preds$M1
ggplot(cal, aes(x = p, y = obs, color = locale)) +
  geom_abline(slope = 1, intercept = 0, linetype = "dashed", color = "grey40") +
  geom_point(alpha = 0.35, size = 0.9) +
  scale_color_brewer(palette = "Set2", name = "Held-out locale") +
  coord_equal(xlim = c(0, 1), ylim = c(0, 1)) +
  labs(
    title = "Held-out calibration (M1 baseline)",
    subtitle = "Each point is a precinct predicted while its locale was held out",
    x = "Predicted RCV support", y = "Observed RCV support"
  ) +
  theme_minimal(base_size = 12)
```



Imputation uncertainty

For M2 / M3, how much does the choice of imputation move the headline numbers? Small spread means the out-of-sample conclusion is not an artifact of any single imputation.

```
imp_spread %>%
  transmute(
    Model = model,
    `Dev-R² mean` = round(dev_r2_mean, 3),
    `Dev-R² SD`   = round(dev_r2_sd, 3),
    `MAE mean (pp)` = round(100 * mae_mean, 2),
    `MAE SD (pp)`   = round(100 * mae_sd, 2),
    `Imputations` = m
  ) %>%
  kable(caption = paste0("Spread of headline metrics across the ",
                         prov$m_imputations,
                         " individual imputations (M1 has none).")) %>%
  kable_styling(full_width = FALSE)
```

Table 4: Spread of headline metrics across the 5 individual imputations (M1 has none).

Model	Dev-R^2 mean	Dev-R^2 SD	MAE mean (pp)	MAE SD (pp)	Imputations
M1	0.569	NA	7.85	NA	1
M2	0.517	0.003	8.70	0.02	5
M3	0.237	0.014	10.73	0.08	5

How to cite these numbers

```
tibble(
  Field = c("CV design", "Folds", "Precincts", "Imputations", "Null model",
            "Pooling", "Generated", "R", "mice", "git"),
  Value = c(prov$cv_design, prov$n_folds, prov$n_precincts, prov$m_imputations,
            prov$null_model, prov$pooling, prov$generated_at, prov$r_version,
            prov$mice_version, prov$git_sha)
) %>%
  kable(caption = "Provenance of the numbers in this report.") %>%
  kable_styling(full_width = FALSE)
```

Table 5: Provenance of the numbers in this report.

Field	Value
CV design	leave-one-locale-out
Folds	8
Precincts	3432
Imputations	5
Null model	intercept-only (training-fold overall support rate)
Pooling	predicted probabilities averaged across imputations, then scored
Generated	2026-07-17 23:39:56 UTC
R	R version 4.6.1 (2026-06-24)
mice	3.19.0
git	319828a

Numbers are persisted to `data/cv_results.RData`, `data/cv_metrics.csv`, and `data/cv_fold_metrics.csv`, regenerated by `R/05_cv.R`.